CS 486/686
Hidden Markov Models II

Yuntian Deng

Lecture 11

RN 14.2.2

Reminders

⏰ **Chat 4 is due TODAY** (Tue Jun 16); Chat 5 due Thu Jun 18 — both 11:59 PM on Chrysalis.

📅 **Assignment 1 deadline extended to Thu Jun 25** (was Jun 18). The PDF, starter code, and model file were **updated on LEARN** — please re-download.

📝 **Assignment 1** questions → **Dake Zhang** on Piazza.

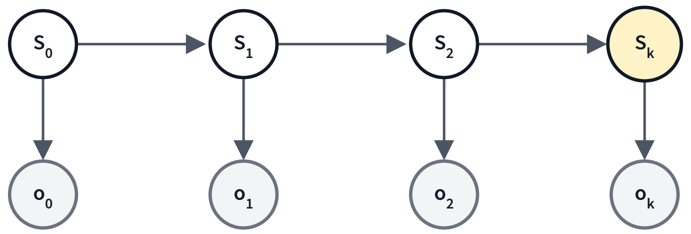
Learning goals

- Compute the **smoothing** probability for a time step in an HMM.
- **Justify** each step in the derivation of the smoothing formula.
- Describe the **forward-backward** algorithm.
- Describe the **Viterbi** algorithm.

Recap: filtering (from L10)

From past evidence $o_{0:k}$, filtering gives the posterior over the *current* state:

$$f_{0:k} = P(S_k|o_{0:k}) = \alpha\, P(o_k|S_k) \sum_{s_{k-1}} P(S_k|s_{k-1})\, f_{0:(k-1)}$$

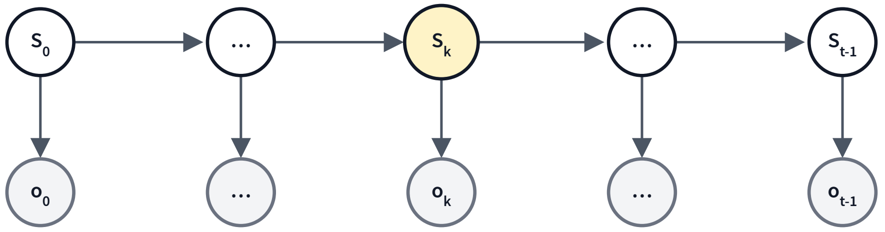


- Yellow = query state S_k (the *present*).
- Gray = observed evidence.
- Cost: $O(k)$ ops via the recursive predict-update.

Today: a query about a **past** state, given all the evidence we have.

Smoothing: query a past state

$$P(S_k | o_{0:(t-1)}), \quad 0 \leq k \leq t-1$$



Filtering (L10)

$P(S_k | o_{0:k})$ — the present state from past evidence.

Smoothing (today)

$P(S_k | o_{0:(t-1)})$, $k < t-1$ — a past state from all evidence (past *and* future).

Deriving the smoothing formula

Four small steps from $P(S_k|o_{0:(t-1)})$ to the forward-backward product:

$$P(S_k|o_{0:(t-1)})$$

$$= P(S_k|o_{(k+1):(t-1)} \wedge o_{0:k}) \qquad \textit{rewrite}$$

$$= \alpha\, P(S_k|o_{0:k})\, P(o_{(k+1):(t-1)}|S_k \wedge o_{0:k}) \qquad \textit{Bayes' rule}$$

$$= \alpha\, P(S_k|o_{0:k})\, P(o_{(k+1):(t-1)}|S_k) \qquad \textit{Markov}$$

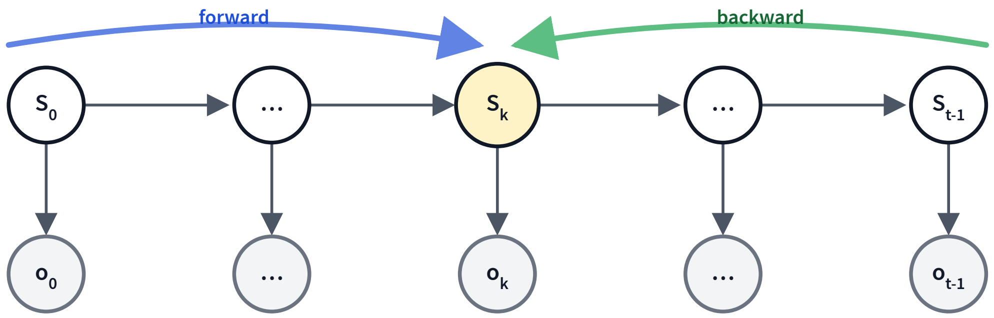
$$= \alpha\, f_{0:k}\, b_{(k+1):(t-1)} \qquad \textit{definitions}$$

Try to identify the rule before the label appears.

Smoothing = forward × backward

Split the evidence around S_k into past and future:

$$P(S_k | o_{0:(t-1)}) = \alpha \underbrace{P(S_k | o_{0:k})}_{\textbf{forward } f_{0:k}} \underbrace{P(o_{(k+1):(t-1)} | S_k)}_{\textbf{backward } b_{(k+1):(t-1)}}$$



Two sweeps meet at S_k; multiply pointwise and normalize.

Backward recursion

Compute $b_{(k+1):(t-1)} \triangleq P(o_{(k+1):(t-1)} | S_k)$ right-to-left.

Base case (k = t − 1, no future observations)

$$b_{t:(t-1)} = \langle 1, 1 \rangle$$

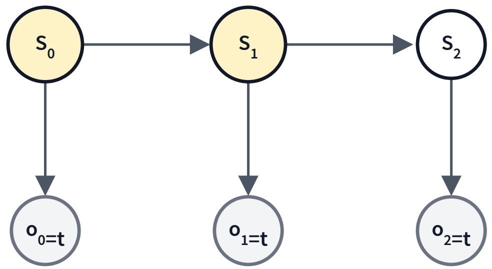
Recursive case (k < t − 1)

$$b_{(k+1):(t-1)} = \sum_{s_{k+1}} P(o_{k+1} \mid s_{k+1})\, b_{(k+2):(t-1)}\, P(s_{k+1} \mid S_k)$$

Same factor sizes as forward recursion — total cost $O(t)$ for the whole backward pass.

No normalization here: b is a likelihood, not a probability. Normalize *after* multiplying $f \times b$.

A smoothing example: setup



Umbrella story, given $O_0 = O_1 = O_2 = t$. Compute the *smoothed* beliefs:

- $P(S_0|o_{0:2})$: did it rain on **day 0**?
- $P(S_1|o_{0:2})$: did it rain on **day 1**?

From L10 we already have $f_{0:0} = \langle 0.818, 0.182 \rangle$ and $f_{0:1} = \langle 0.883, 0.117 \rangle$.

Backward pass: compute $b_{2:2}$, then $b_{1:2}$

$b_{2:2} = P(o_2|S_1) = \sum_{s_2} P(o_2|s_2)\, b_{3:2}\, P(s_2|S_1)$ with $b_{3:2} = \langle 1, 1\rangle$:

$$= 0.9 \cdot 1 \cdot \langle 0.7, 0.3\rangle + 0.2 \cdot 1 \cdot \langle 0.3, 0.7\rangle$$

$$= \langle 0.63, 0.27\rangle + \langle 0.06, 0.14\rangle = \langle \mathbf{0.69}, \mathbf{0.41}\rangle$$

$b_{1:2} = P(o_{1:2}|S_0) = \sum_{s_1} P(o_1|s_1)\, b_{2:2}\, P(s_1|S_0)$:

$$= 0.9 \cdot 0.69 \cdot \langle 0.7, 0.3\rangle + 0.2 \cdot 0.41 \cdot \langle 0.3, 0.7\rangle$$

$$= \langle 0.4347, 0.1863\rangle + \langle 0.0246, 0.0574\rangle = \langle \mathbf{0.4593}, \mathbf{0.2437}\rangle$$

Combine: $P(S_k | o_{0:2}) = \alpha\, f_{0:k}\, b_{(k+1):2}$

$$P(S_1 | o_{0:2}) = \alpha\, f_{0:1} \cdot b_{2:2}:$$

$$= \alpha\, \langle 0.883, 0.117 \rangle \cdot \langle 0.69, 0.41 \rangle = \alpha\, \langle 0.6093, 0.0480 \rangle = \langle \mathbf{0.927}, \mathbf{0.073} \rangle$$

$$P(S_0 | o_{0:2}) = \alpha\, f_{0:0} \cdot b_{1:2}:$$

$$= \alpha\, \langle 0.818, 0.182 \rangle \cdot \langle 0.4593, 0.2437 \rangle = \alpha\, \langle 0.3757, 0.0444 \rangle = \langle \mathbf{0.894}, \mathbf{0.106} \rangle$$

	$P(S_1 = T \mid \ldots)$
Filtering with $o_{0:1}$ only	0.883
Smoothing with $o_{0:2}$	**0.927**

The extra "umbrella on day 2" makes "rain on day 1" more likely — future evidence helps.

Deriving the backward recursion

Five steps from $P(o_{(k+1):(t-1)}|S_k)$ to the recursive form:

$$P(o_{(k+1):(t-1)}|S_k)$$

$$= \sum_{s_{k+1}} P(o_{(k+1):(t-1)} \wedge s_{k+1}|S_k)$$ *sum rule*

$$= \sum_{s_{k+1}} P(o_{(k+1):(t-1)}|s_{k+1} \wedge S_k)\, P(s_{k+1}|S_k)$$ *chain rule*

$$= \sum_{s_{k+1}} P(o_{(k+1):(t-1)}|s_{k+1})\, P(s_{k+1}|S_k)$$ *Markov*

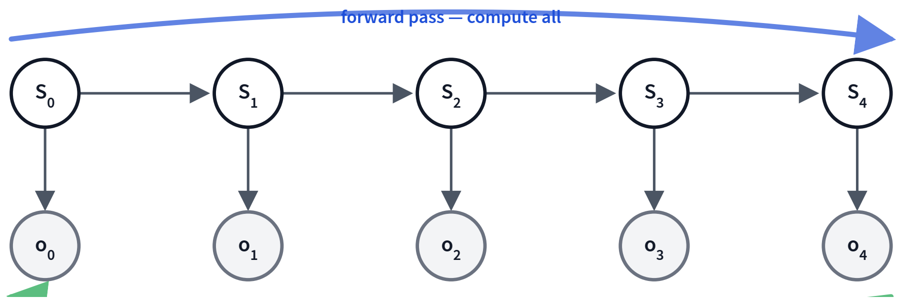
$$= \sum_{s_{k+1}} P(o_{k+1} \wedge o_{(k+2):(t-1)}|s_{k+1})\, P(s_{k+1}|S_k)$$ *rewrite*

$$= \sum_{s_{k+1}} P(o_{k+1}|s_{k+1})\, P(o_{(k+2):(t-1)}|s_{k+1})\, P(s_{k+1}|S_k)$$ *Markov*

The final factor is exactly $P(o_{k+1}|s_{k+1})\, b_{(k+2):(t-1)}\, P(s_{k+1}|S_k)$ — the backward recursion.

The forward-backward algorithm

Compute the smoothed posterior at *every* time step k with one forward sweep + one backward sweep.



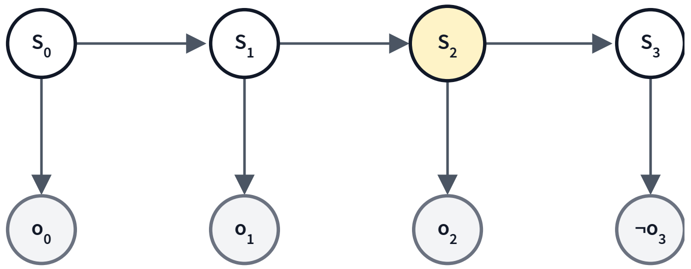
Combine the two sweeps at every k

$$P(S_k \mid o_{0:(t-1)}) = \alpha\, f_{0:k} \cdot b_{(k+1):(t-1)}$$

Forward gives every $f_{0:k}$ (recap of L10), backward gives every $b_{(k+1):(t-1)}$ — total cost $O(t)$, one sweep each direction.

Try it: smoothing on a 4-step HMM

Compute $P(S_2 \mid o_0 \wedge o_1 \wedge o_2 \wedge \neg o_3)$ – here $k = 2,\ t = 4$.



Numbers

$P(s_0) = 0.4$
$P(s_t|s_{t-1}) = 0.7$
$P(s_t|\neg s_{t-1}) = 0.2$
$P(o_t|s_t) = 0.9$
$P(o_t|\neg s_t) = 0.2$

Forward (from L10): $f_{0:2} \approx \langle 0.884, 0.116\rangle$

Backward: $b_{3:3} = 0.1\langle 0.7, 0.2\rangle + 0.8\langle 0.3, 0.8\rangle = \langle 0.31, 0.66\rangle$

Smoothed: $\alpha\langle 0.884, 0.116\rangle \cdot \langle 0.31, 0.66\rangle = \alpha\langle 0.274, 0.0766\rangle = \boxed{\langle 0.782, 0.218\rangle}$

The most-likely explanation

Find the single **best sequence of hidden states** that explains the observations:

$$\hat{s}_{0:t-1} = \arg\max_{s_{0:t-1}} P(s_{0:t-1} | o_{0:t-1})$$

- Generalize: $S_t \in \{0, 1, \ldots, n-1\}$ — not just binary.
- Transition matrix $A \in \mathbb{R}^{n \times n}$, emission matrix $O \in \mathbb{R}^{n \times m}$.

Brute force: enumerate every state sequence.

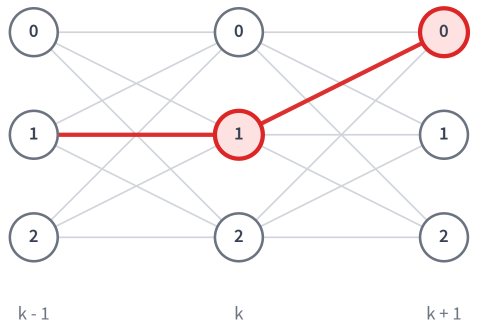
n^t sequences — intractable.

Dynamic programming to the rescue: the **Viterbi algorithm**.

Dynamic programming: define $\pi_k(j)$

$$\pi_k(j) \;=\; \max_{s_{0:(k-1)}} P\big(s_{0:(k-1)},\, S_k = j \mid o_{0:k}\big)$$

The highest probability of any state sequence $s_{0:k}$ ending with $S_k = j$, given observations $o_{0:k}$.



Trellis: one column per time step, one node per state value. Red = best path entering each node.

Why the recursion holds

The best path to state j at time k reuses the best path to some predecessor z at $k-1$
:

$$\pi_k(j) = \max_{s_{0:(k-1)}} P(s_{0:(k-1)},\, S_k{=}j \mid o_{0:k})$$

$$\propto \max_{s_{0:(k-1)}} P(s_{0:(k-1)},\, S_k{=}j,\, o_{0:k}) \qquad \textit{drop } P(o_{0:k})$$

$$= \max_{s_{0:(k-1)}} P(s_{0:(k-1)},\, o_{0:(k-1)})\ \underbrace{P(S_k{=}j \mid s_{k-1})}_{\text{transition}}\ \underbrace{P(o_k \mid j)}_{\text{emission}} \qquad \textit{HMM factorization}$$

$$= P(o_k \mid j) \max_{s_{k-1}} \max_{s_{0:(k-2)}} P(s_{0:(k-2)},\, s_{k-1},\, o_{0:(k-1)})\, P(j \mid s_{k-1}) \qquad \textit{nest the prefix max}$$

$$= P(o_k \mid j) \max_{z} P(j \mid z) \max_{s_{0:(k-2)}} P(s_{0:(k-2)},\, S_{k-1}{=}z,\, o_{0:(k-1)}) \qquad \textit{pull out } P(j \mid z)$$

$$= P(o_k \mid j) \max_{z} \big[\, P(j \mid z)\, \pi_{k-1}(z) \,\big] \qquad \textit{inner max} = \pi_{k-1}(z)$$

The inner max has no j in it — so **one best path per node** serves every successor ($\phi_k(j)$ records the winning z). Constants don't change the argmax, so we work with the joint.

Viterbi recursion

Base case

$$\pi_0(j) = \alpha\, P(S_0 = j)\, P(o_0|S_0 = j)$$

Recursive case

$$\pi_k(j) = P(o_k|S_k = j) \cdot \max_z \left[\pi_{k-1}(z)\, P(S_k = j|S_{k-1} = z)\right]$$

Backpointer

$$\phi_k(j) = \arg\max_z \left[\pi_{k-1}(z)\, P(S_k = j|S_{k-1} = z)\right]$$

$\pi_k(j)$ = best probability of any path ending at state j at time k. $\phi_k(j)$ remembers which predecessor produced it — we'll use it to trace the best sequence back.

Viterbi: the algorithm

`viterbi(observations, transition, emission)`

1. Initialize $\pi_0(j) = \alpha\, P(S_0 = j)\, P(o_0 | S_0 = j)$ for each state j; $\phi_0(j) = \text{none}$.
2. **Forward sweep**: for $k = 1, \ldots, t-1$ and each j, compute $\pi_k(j)$ and record the argmax in $\phi_k(j)$.
3. **Best terminal**: $\hat{s}_{t-1} = \arg\max_j \pi_{t-1}(j)$.
4. **Traceback**: for $k = t-1, \ldots, 1$, set $\hat{s}_{k-1} = \phi_k(\hat{s}_k)$.
5. Return the sequence $(\hat{s}_0, \hat{s}_1, \ldots, \hat{s}_{t-1})$.

One sweep fills the trellis; one walk back along the backpointers extracts the MAP sequence.

Why Viterbi returns the true argmax

Recall $\pi_k(j) = \max\limits_{s_{0:(k-1)}} P(s_{0:(k-1)}, S_k{=}j, o_{0:k})$ and $\phi_k(j) = \arg\max_z \left[\pi_{k-1}(z)\, P(j \mid z)\right]$. Goal: $\hat{s}_{0:t-1} = \arg\max_{s_{0:t-1}} P(s_{0:t-1}, o_{0:t-1})$.

$\hat{s}_{t-1}, \hat{s}_{t-2}, \ldots, \hat{s}_0 = \arg\max\limits_{s_{0:t-1}} P(s_{0:t-1}, o_{0:t-1})$ *the MAP path*

$= \arg\max\limits_{s_{t-1}} \arg\max\limits_{s_{0:t-2}} P(s_{0:t-1}, o_{0:t-1})$ *split: pick s_{t-1}, then the rest*

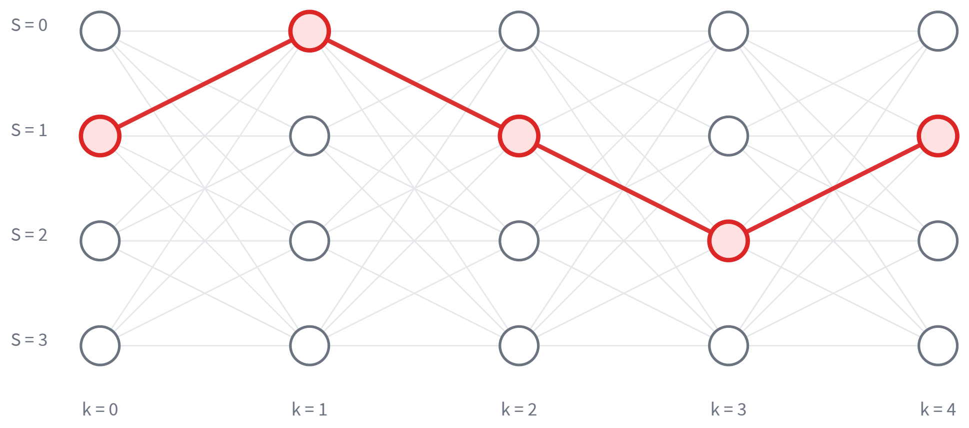
$\hat{s}_{t-1} = \arg\max\limits_{s_{t-1}} \max\limits_{s_{0:t-2}} P(s_{0:t-1}, o_{0:t-1}) = \arg\max\limits_{s_{t-1}} \pi_{t-1}(s_{t-1})$ *outer argmax = best final state*

$\hat{s}_{t-2} = \arg\max\limits_{s_{t-2}} \pi_{t-2}(s_{t-2})\, P(\hat{s}_{t-1} \mid s_{t-2}) = \phi_{t-1}(\hat{s}_{t-1})$ *extract one more (drop const. emission)*

$\hat{s}_{k-1} = \phi_k(\hat{s}_k), \quad k = t{-}1, \ldots, 1$ *repeat down the chain* ■

Splitting the argmax one state at a time turns the global search into t tiny argmaxes — and each one is exactly a backpointer ϕ_k.

Viterbi: trellis walk-through + complexity



Best path: $\hat{s} = (1, 0, 1, 2, 1)$.

Complexity: $O(t \cdot n^2)$ — one trellis edge per cell per step. Linear in time, quadratic in #states.

Learning goals (recap)

✓ Compute the **smoothing** probability for a time step in an HMM.

✓ **Justify** each step in the smoothing & backward derivations.

✓ Describe the **forward-backward** algorithm.

✓ Describe the **Viterbi** algorithm.

Search › Uncertainty › **Decisions** › Learning

Next module: Decision Making

So far we have *beliefs*. Next we add *preferences* — how should an agent **act** when outcomes are uncertain?

- **L12** — decision theory: utility, MEU, decision networks, value of information.
- **L13–L14** — Markov decision processes: Bellman equation, value & policy iteration.
- **L15** — reinforcement learning: ADP, Q-learning, SARSA.